

Exercise Sheet 1

Exercise 1. $\pi(x) \leq \log \log x$. pf: $p_1 \leq p_2 \leq \dots \leq p_n \leq x$, 归纳, $p_n \leq 2^{n-1} = e^{n-1}$, 取对数.
Exercise 2. (a) $(n)^2 = \sum_{d|n} \mu(d) \tau(n/d)$; (b) $\sum_{d|n} \mu(d) \tau(n/d) = 1$ k 次因式 F; (c) $\sum_{1 \leq n \leq x, (a,n)=1} e^{2\pi i a/n}$. (a) pf: $\tau^2 = \mu * \tau$, 只需 $\tau^2 = \tau^2 * \mu$, 显. (b) pf: 最大 k 次因式设为 $p_1^{k_1} \dots p_m^{k_m}$, 若为 1, 则 LHS=1, 否则为 $\sum_{d|p_1 \dots p_m} \mu(d) = 0$. (c) pf: 先证 RHS 可乘, 因为 m, n 互素时, 在模 mn 意义下 $\{am + bn | a \in (\mathbb{Z}/n)^{\times}, b \in (\mathbb{Z}/m)^{\times}\} = (\mathbb{Z}/mn)^{\times}$, 再说 p^n 时 LHS=RHS (用 p 进制).
Exercise 3. (a) $\sum_{d|n} \mu(d)^2 = 2^{\omega(n)}$; (b) $\sum_{d|n} \mu(d) \tau(d) = (-1)^{\omega(n)}$. (a) pf: 显. (b) pf: LHS = $\sum_{k=0}^{\omega(n)} \binom{\omega(n)}{k} (-1)^k 2^k = (1-2)^{\omega(n)} = (-1)^{\omega(n)}$.
Exercise 4. f : 可乘, 则 f^{-1} 完全可乘 iff $f(p^2) = 0$, $\ell \geq 2$. pf: 已知可乘则完全可乘等价于 $f^{-1} = f \mu$ 故显.
Exercise 5. f : $f(p) = f(u)$ 恰有两个, 是否可乘? pf: f 满足 $f(n) = \pm 1$, 选准 $f(n) = \frac{1}{27171} (1 - \sum_{d|n, d \neq 1, n} f(d) f(n/d))$, 该选准确定, 故恰有两个. 若 $f(1) = -1$ 不可乘; $f(1) = 1$ 可乘 (用反证法找最小的 mn).
Exercise 6. $(f * g)^{-1} = f^{-1} * g^{-1}$. 显.
Exercise 7. $\text{supp } \Lambda_k \subseteq \{n: \omega(n) \leq k\}$; $0 \leq \Lambda_k(n) \leq (\log n)^k$. ① pf: $\Lambda_k = \Lambda_{k-1} \log + \Lambda_{k-1} * \Lambda$, 故显, ② $\Lambda \geq 0$, 是因为递推式, “ $\leq$ ” 是因为 $\log^k * \mu * u \Rightarrow \Lambda_k(n) = \log^k n - \sum_{d|n, d < n} \Lambda_k(d) \leq \log^k n$.

Exercise Sheet 2

Exercise 1. 求 $f \in A$ s.t. $\frac{f(n)}{\varphi(n)} = \sum_{d|n} \frac{1}{d} f(n/d)$. pf: $a(n) = 1/n$ 有 $a^{-1}(n) = \mu(n)/n$, 故 $f = \frac{1}{\varphi} * a^{-1}$ 可乘, $f(p) = \frac{1}{p(p-1)} f(p^0) = 0$ ($\alpha \geq 2$), 故 $f(n) = \mu(n)^2 \prod_{p|n} \frac{1}{p-1} = \frac{\mu(n)^2}{n \varphi(n)}$.

Exercise 2. $\sum_{n \leq x} \frac{f(n)}{p(n-1)} = C_1 \log x + C_2 + O(\log x/x)$. pf: $\sum_{n \leq x} \frac{1}{p(n-1)} = \sum_{n \leq x} a * f(n) = \sum_{d \leq x} f(d) \sum_{m \leq x/d} \frac{1}{m} = \sum_{d \leq x} f(d) (\log \frac{x}{d} + \gamma + O(d/x)) := \sum_{d \leq x} f(d) (\log x - (\sum_{d \leq x} f(d) \log d + \gamma \sum_{d \leq x} f(d)) - \log x \sum_{d \leq x} f(d) - (\sum_{d \leq x} f(d) \log d + \gamma \sum_{d \leq x} f(d)) + O(\sum_{d \leq x} f(d) d/x))$, 由以下估计可证原命题: ① $\sum_{d \geq 2} f(d) \log d$ 收敛, pf: $\varphi(n) \gg n/\log \log n$ (由此也可知 $\sum_{n \leq x} 1/\varphi(n) = O(\log x)$), ② $\sum_{d \leq x} f(d) \log d = O(\log x)$, pf: $|\sum_{d \leq x} f(d) \log d| \leq |\sum_{d \leq x} \mu(d)^2 (\log d)/\varphi(d)| \leq |\sum_{d \leq x} 1/\varphi(d)| = O(\log x)$, ③ $\sum_{d > x} f(d) \log d = O(1/x)$, pf: $f(d) = d^{-2} \mu(d)^2 / \varphi(d)$, 记为 $d^{-2} b(d)$, 由 2-1, $\frac{\mu(n)^2}{n} = n * f(n) = n \sum_{d|n} \frac{\mu(d)^2}{d} = \sum_{d|n} \frac{\mu(d)^2}{d}$, 即 $|b(n)| \leq n/\varphi(n) = \sum_{d|n} \mu(d)^2 / \varphi(d)$, 那 $|\sum_{n \leq x} b(n)| \leq \sum_{n \leq x} \sum_{d|n} \mu(d)^2 / \varphi(d) = \sum_{d \leq x} \frac{\mu(d)^2}{\varphi(d)} [x/d] \leq x \sum_{d \leq x} \frac{\mu(d)^2}{\varphi(d)} = x \sum_{d \leq x} f(d) = O(x)$, 即 $\sum_{d > x} f(d) = \lim_{n \rightarrow \infty} \sum_{x < d \leq n} b(d) \log d = \lim_{n \rightarrow \infty} \sum_{x < d \leq n} \frac{1}{d^2} d \log d \leq b(d) = \lim_{n \rightarrow \infty} O(1/N) - 2 \int_x^N \xi^{-3} \sum_{x < d \leq \xi} b(d) d \log \xi = 2 \int_x^N \xi^{-3} O(\log \xi) d\xi = O(1/x)$, ④ $\sum_{d > x} f(d) \log d = O(\log x/x)$, pf: $\text{LHS} = \sum_{d > x} b(d) \log d \log d = \sum_{x < d \leq n} \sum_{x < d \leq \xi} b(d) \log d \log d = \lim_{n \rightarrow \infty} \sum_{x < d \leq \xi} \frac{1}{\log \xi} b(d) d \log d = O(\log N/N) - \frac{1}{N} \sum_{x < d \leq \xi} b(d) d \log d = 2 \int_x^N \frac{1}{\log \xi} \sum_{x < d \leq \xi} b(d) d \log \xi = 2 \int_x^N \frac{1}{\log \xi} d\xi = O(\log x/x)$ 注意 $\log x + 1/x$ 导数是 $-\log x/x^2$, 故得证.

Exercise 3. 对 $x > 3$, $\sum_{1 \leq n \leq x} \frac{1}{n \log x} = \log \log x + C_1 + O(1/(x \log x))$, pf: Euler-Maclaurin, $\text{LHS} = \int_2^x \frac{d\xi}{\xi^2 \log \xi} + \int_2^x \frac{1}{\xi^2} (\xi - \frac{1}{2}) (\frac{1}{\xi \log \xi})' d\xi + ((x - \frac{1}{2}) \frac{1}{x \log x} - (\{2\} - \frac{1}{2}) \frac{1}{2 \log 2}) = (\log \log x - \log \log 2 + \frac{1}{2 \log 2} + \int_2^x \{ \xi \} - \frac{1}{2} (\frac{1}{\xi \log \xi})' d\xi + ((x - \frac{1}{2}) \frac{1}{x \log x} - (\{2\} - \frac{1}{2}) \frac{1}{2 \log 2}))$, 即所求.

4 $\sum_{0 \leq a \leq \sqrt{x}} \sqrt{x-a^2} + O(\sqrt{x})$. 关于 a 单调, $\sum_{0 \leq a \leq \sqrt{x}} \sqrt{x-a^2} = \int_0^{\sqrt{x}} \sqrt{x-a^2} da + O(\sqrt{x}) = -\frac{x \arccos(a/\sqrt{x})}{2} \Big|_0^{\sqrt{x}} + \frac{a\sqrt{x-a^2}}{2} \Big|_0^{\sqrt{x}} + O(\sqrt{x}) = \frac{\pi}{4}x + O(\sqrt{x})$, 且 $a=0$ 值也

Exercise 5. $Q_2(x) = \#\{n \leq x \mid n \text{ 无平方因子}\}$, 则 $Q_2(x) = C_2x + O(\sqrt{x})$ ($C_2 = 6/\pi^2$). pf: $Q_2(x) = \sum_{n \leq x} \mu^2(n) = \sum_{n \leq x} \sum_{d^2 \mid n} \mu(d) = \sum_{d \leq \sqrt{x}} [x/d^2] \mu(d) = \sum_{d \leq \sqrt{x}} \frac{x\mu(d)}{d^2} + O(\sqrt{x}) = x \sum_{d \leq 1} \frac{\mu(d)}{d^2} + x \sum_{d > \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x}) = \frac{1}{\zeta(2)}x + O(x \int_{\sqrt{x}}^{\infty} \xi^{-2} d\xi) + O(\sqrt{x}) = \frac{1}{\zeta(2)}x + O(\sqrt{x})$.

Exercise 6. $\forall \varepsilon > 0, \exists C_\varepsilon > 0, \tau(n) < C_\varepsilon n^\varepsilon$. pf: 选 P s.t. $\forall p > P, \forall \alpha \in \mathbb{N}_{\geq 1}, \alpha+1 < p^{\varepsilon\alpha}$, 再取 $C_\varepsilon = \prod_{p < P} \sup_{\alpha \geq 1} (\alpha+1)/p^{\varepsilon\alpha} < \infty$, 则 $\tau(\prod p_i^{\alpha_i}) = \prod (\alpha_i+1) \leq C_\varepsilon \prod p_i^{\varepsilon\alpha_i}$.

Exercise Sheet 3

Exercise 1. 若 $\varphi_{2^k} \leq f(p) \log p = (a+b) \log x + cx + O(1)$ ($x \geq 2$), 则 $\mathcal{B}D = B(f)$, $\varphi_{2^k} \leq f(p) = ax + (a+c)(\frac{a+b}{\log x} + j_2 \frac{1}{\log^2 x}) + b \log \log x + b + O(\frac{1}{\log x})$. 而 $E(x) = \sum_{p \leq x} f(p) \log p \log \log p$, LHS $= \sum_{2^k \leq n \leq 2x} f(n) \log n \delta_p(n) \frac{\text{Abel}}{n \log x} = \frac{1}{f_3/2} \frac{\text{Abel}}{\log x} + f_3/2 \cdot A(\xi) \int_{\log x}^x \log t \, dt = [\text{案件}] ax + b + c \frac{a+b}{\log x} + O(\frac{1}{\log x}) + f_3/2 \cdot \frac{a+b+c}{\xi \log x} + \frac{c}{\log x} + \frac{E(\xi)}{\xi \log^2 x}$, 积分, 其中 $E(\xi) \ll 1$, 注意 $\frac{1}{x \log x} = \log \log x$ 及 $\int_2^x \frac{d\log x}{\log x} = \frac{1}{2} \log^2 x$, $\int_2^x \frac{d\log x}{\log^2 x} = \frac{1}{2} \log x$, 积分为 0 调整后放在 B 中.

Exercise 2. 证明对 $n \geq 16$, $\sum_{p|n} \frac{1}{p} \ll \log \log \log n$. pf: $\sum_{p|n} \frac{1}{p} = \sum_{p|n, p \leq \log n} \frac{1}{p} + \sum_{p|n, p > \log n} \frac{1}{p} \ll$ (by Mertens) $\log \log \log n + \sum_{p|n, p > \log n} \frac{1}{p}$. 由于 $(\log n)^{\log n / \log \log n} = n$ 从而 $> \log n$ 素数个数不超过 $\frac{\log n}{\log \log n}$, 于是 $\sum_{p|n, p > \log n} \frac{1}{p} \leq \frac{1}{\log \log n} \cdot \frac{\log n}{\log \log n} \ll \log \log \log n$.

Exercise 3. 证明 $\sum_p \frac{\log \log p}{p(\log \log p)^\alpha}$, $\alpha > 1$ 收敛, $\alpha \leq 1$ 发散. pf: 忽略前面的项, $= \lim_{n \rightarrow \infty} S_n$, 其中 $S_n = \sum_{5 < n \leq N} \delta_p(n) \frac{1}{n} \frac{1}{(\log \log n)^\alpha} = \sum_{5 < p \leq N} \frac{1}{p} \frac{1}{(\log \log N)^\alpha} + \int_5^N \sum_{5 < p \leq x} \frac{1}{p} \frac{1}{(\log \log x)^\alpha} + 1 \frac{dx}{x \log x}$, 而 $\sum_{5 < p \leq x} \frac{1}{p} \sim \log \log x$ 且 $\int_5^N \frac{dx}{x \log x (\log \log x)^\alpha}$ 在 $\alpha > 1$ 收敛, $\alpha \leq 1$ 发散 (用换元 $u = \log \log x$, $\frac{dx}{x \log x} = du$) 即证.

Exercise 4. 设 $S(x) = \#\{n \leq x \mid p \mid n \Rightarrow p \leq \sqrt{n}\}$, 证明 $S(x) = (1 - \log 2)x + O(x/\log x)$. pf: $S(x) = [x] - \sum_{\sqrt{x} < p \leq x} \sum_{m \leq x/p} 1 = [x] - \sum_{\sqrt{x} < p \leq x} \sum_{m \leq x/p} 1 = [x] - \sum_{\sqrt{x} < p \leq x} \sum_{m \leq \sqrt{x}} 1 - \sum_{\sqrt{x} < p \leq x} \sum_{\sqrt{x} < m \leq x/p} 1$, 而 $[x] = x + O(1)$, $\sum_{\sqrt{x} < p \leq x} (p-1) \leq \sqrt{x} \pi(\sqrt{x}) = O(x/\log x)$, $\sum_{\sqrt{x} < p \leq x} \sum_{\sqrt{x} < m \leq x/p} 1 = x(\log \log x) + C + O(\frac{x}{\log x}) - (\log \log \sqrt{x} + C + O(\frac{\sqrt{x}}{\log \sqrt{x}})) = O(x/\log x)$, 代入即得.

Exercise 5. 若 $\sum_{n \leq x} \frac{\mu(n)}{n} = o(1)$, 则 PNT. pf: $\sum_{1/2 < n \leq x} \mu(n) = \sum_{1/2 < n \leq x} \frac{\mu(n)}{n} n \stackrel{\text{Abel}}{=} x \sum_{1/2 < n \leq x} \frac{\mu(n)}{n} - \int_{1/2}^x \sum_{1/2 < n \leq t} \frac{\mu(n)}{n} dt = o(x) + \int_{1/2}^x M dt + \int_{1/2}^x o(1) dt = o(x) \Rightarrow \text{PNT}.$

Exercise 6. $L(N) = \text{lcm}\{1, 2, \dots, N\}$, 求证 $\text{PNT} \Leftrightarrow \lim_{N \rightarrow \infty} L(N)^{1/N} = e$. pf: $\log L(N) = \sum_{p^a \leq N} \log p = \psi(N)$, 故 $L(N)^{1/N} \rightarrow e$ iff $\frac{\psi(N)}{N} \rightarrow 1$ iff $\psi(x) \sim x$ iff PNT.

Exercise 7. $\forall n > 1, \exists p$ s.t. $p \parallel n$. pf: Bertrand's 假设, 取 $\frac{n}{2} < p < n$, 则 $2p > n$, 之后显然.

Exercise Sheet 4

Exercise 1. TFAE: (a) $f(n) \ll n^A (A > 0)$; (b) $\sigma_a < +\infty$. pf: ① (a) 蕴含 $|f(n)| \leq Cn^A$, 故 $|L_f(s)| \leq \sum_{n \geq 1} \frac{C}{n^{\sigma-A}}$, 即 $\text{Re}(s) > A+1$ 时 $L_f(s)$ 绝对收敛, 那 $\sigma_a < +\infty$.
② (b) 蕴含 $\exists c > 0, \sum_{n=1}^{\infty} \frac{f(n)}{n^c}$ 绝对收敛, 那 $\lim_{n \rightarrow \infty} \frac{|f(n)|}{n^c} = 0$, 则 $f(n) \ll n^c$.

Exercise 2. (a) 找例

Exercise 3. 找 $F(s) = \sum_{m_1, m_2, m_3 \geq 1} (m_1 + m_2 + m_3)s^{-(m_1 + m_2 + m_3)}$ 的绝对收敛横坐标, 用 ζ 表示. $F(s) = \sum_{n \geq 1} \sum_{m_1 + m_2 + m_3 = n} n^{-s} = \sum_{n \geq 1} \binom{n-1}{2} n^{-s} = \sum_{n \geq 1} \frac{(n-1)(n-2)}{2} n^{-s} = \frac{1}{2} \zeta(s-2) - \frac{1}{2} \zeta(s-1) + \frac{1}{2} \zeta(s)$ ($n = 0, 1, 2$ 项抵消).

Exercise 4. $\chi(n)$ 定义为 $\chi(1) = 1, \chi(p_1^{a_1} \cdots p_k^{a_k}) = \alpha_1 \alpha_2 \cdots \alpha_k$, 计算 (a) $\sum_{n \geq 1} \frac{\tau(n^2)}{n^s}$; (b) $\sum_{n \geq 1} \frac{2^{\omega(n)}}{n^s}$; (c) $\sum_{n \geq 1} \frac{(-1)^{\Omega(n)}}{n^s}$; (d) $\sum_{n \geq 1} \frac{3^{\omega(n)} \chi(n)}{n^s}$. pf: (a) $\tau(n^2)$ 可乘, 用 Euler 乘积, $\sum_{n \geq 1} \frac{\tau(n^2)}{n^s} = \prod_p (1 + \sum_{\alpha \geq 1} \frac{\tau(p^{2\alpha})}{p^{s\alpha}}) = \prod_p (1 + \sum_{\alpha \geq 1} \frac{2+1}{p^{s\alpha}}) = \prod_p ((1 - p^{-s})^{-1} (1 - p^{-2s})^{-1} (1 + p^{-s})) = \prod_p (1 - p^{-s})^{-3} (1 - p^{-2s}) = \frac{\zeta(s)^3}{\zeta(2s)}$, (b) $2^{\omega(n)}$ 可乘, 原式 $= \prod_p (1 + \sum_{\alpha \geq 1} \frac{2}{p^{s\alpha}}) = \prod_p (1 - p^{-2s}) \prod_p (1 - p^{-s})^{-2} = \frac{\zeta(s)^2}{\zeta(2s)}$, (c) 显然 $= \frac{\zeta(2s)}{\zeta(s)}$ (是 Liouville 函数), (d) 可乘, $= \prod_p (1 + \sum_{\alpha \geq 1} \frac{3^\alpha}{p^{s\alpha}})$ (错位相减) $= \prod_p (1 + \frac{3p^s}{(p^s-1)^2}) = \prod_p (1 - p^{-3s})(1 - p^{-s})^{-3} = \frac{\zeta(s)^3}{\zeta(3s)}$.

Exercise 5. For $c > 0$, $f_c = \frac{c}{c-1} \int_{c-1}^{+\infty}$. 证明 $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} ds = 0$ if $0 < y \leq 1$, $= \log y$ if $y \geq 1$; 用包含 Λ 的和式表示 $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} (-\zeta'/\zeta)(s) ds$ Solution. 模仿讲义图道积分, 模仿讲义 peron 公式可得 $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} (-\zeta'/\zeta)(s) ds = \sum_{n \leq y} \Lambda(n) \log(y/n)$

Exercise 6. $\operatorname{Re}(s) > 1, \frac{1}{\Gamma(s)} = s \int_0^\infty \frac{M(x)}{x^{s+1}} dx, -\frac{\zeta(s)}{\Gamma(s)} = s \int_1^\infty \frac{\psi(x)}{x^{s+1}} dx$, 其中 $M(x) = \sum_{n \leq x} \mu(n), \psi(x) = \sum_{n \leq x} \Lambda(n)$. pf: (a) $s \int_0^\infty \frac{M(x)}{x^{s+1}} dx = s \sum_{n=1}^\infty \mu(n) \int_n^\infty \frac{1}{x^{s+1}} dx = -\left[\sum_{n=1}^\infty \mu(n) \frac{1}{n^s} - \sum_{n=1}^\infty \mu(n) \frac{1}{n^s} \right]$ (取 $N \rightarrow \infty$) $= \sum_{n=1}^\infty \mu(n) \frac{1}{n^s} = -\frac{\zeta(s)}{\Gamma(s)}$. (b) $s \int_1^\infty \frac{\psi(x)}{x^{s+1}} dx = s \sum_{n=1}^\infty \Lambda(n) \int_n^\infty \frac{1}{x^{s+1}} dx = -\left[\sum_{n=1}^\infty \Lambda(n) \frac{1}{n^s} - \sum_{n=1}^\infty \Lambda(n) \frac{1}{n^s} \right]$

Exercise Sheet 5

Exercise 1. $k = 2^a$, $\forall a$ 奇, $a \equiv (-1)^{-1-1} 5^0 7^0 (a) \pmod{2^a}$, 对 $a \geq 2$, $(a-1)a \equiv \frac{a-1}{2} \pmod{2}$; (b) $a \equiv a' \pmod{2^a}$ 则 $r_{-1}(a') \equiv r_{-1}(a) \pmod{2}$, $r_0(a') \equiv r_0(a) \pmod{2^{a-2}}$.
Exercise 2. pf: 显见. 讲义.

Exercise 3. $\varphi: K \rightarrow K$ 是 K 上非平凡自同构, 证明: (a) φ 是 K 上非平凡 $\mathbb{Q}$ -线性映射, 且 $\varphi(x) = x$ 当且仅当 $x \in \mathbb{Q}$; (b) φ 是 K 上非平凡 $\mathbb{Q}$ -线性映射, 且 $\varphi(x) = x$ 当且仅当 $x \in \mathbb{Q}$; (c) φ 是 K 上非平凡 $\mathbb{Q}$ -线性映射, 且 $\varphi(x) = x$ 当且仅当 $x \in \mathbb{Q}$.

Exercise 4. $k = 2^\alpha$, $\alpha \geq 3$, $\chi(n, k, m_{-1}, m_0) = (-1)^{m-1r-1(n)} e^{\frac{m_0 r_0(n)}{2^\alpha - 2}}$, $((n, 2) = 1)$, 本原 iff $(m_0, 2) = 1$. pf: 见 6.3.6.

Exercise 5. 构造 4 非主特征底原: $\chi_1(n) = (-1)^{\frac{n-1}{2}}$, $((n, 2) = 1)$. 构造 8 底原主特征底原: $\chi_2(n) = (-1)^{\frac{n-1}{2}}$, $((n, 2) = 1)$, $\chi_3(n) \chi_2(n) = (-1)^{\frac{(n-1)(n+5)}{8}}$, $((n, 2) = 1)$. pf: 查

接计算。

Exercise 6. $f: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$, $f: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}(\alpha) := \frac{1}{q} \sum_{f \in \mathbb{Z}/q\mathbb{Z}} f(a) e^{2\pi i a \alpha / q}$, $(a, f, g: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C})$ 证明 $\sum_{f \in \mathbb{Z}/q\mathbb{Z}} f(a) \overline{g(a)} = \sum_{a \in \mathbb{Z}/q\mathbb{Z}} f(a) \overline{g(a)}$; $(f) \sum_{a \in \mathbb{Z}/q\mathbb{Z}} |f(a)|^2$ $\sum_{a \in \mathbb{Z}/q\mathbb{Z}} |\widehat{f}(\alpha)|^2$; $C = I(A, B) \subset [0, q]$, 计算 $\widehat{I}(\alpha)$; (d) 求证 $\sum_{a \in \mathbb{Z}/q\mathbb{Z}} |\widehat{I}(\alpha)| \ll \sqrt{q \log q}$; (e) 证明对 A 特征 χ mod q , $\forall N \geq 1$, $\sum_{n=1}^N |\chi(n)| \ll \sqrt{q \log q}$. (a) pf: 换序求和, 显. (用加数特征正交性.) (b) pf: α 中取 f . (c) pf: $\widehat{I}(\alpha) = \frac{1}{q} \sum_{a \in \mathbb{Z}/q\mathbb{Z}} e^{2\pi i a \alpha / q} \ll e^{\frac{2\pi i a \alpha}{q}} \ll e^{\frac{2\pi i a \alpha}{q}}$ (等比数求和). 不假设 $A, B \in \mathbb{Z}$, (否则考虑 $[A+1, |B|]$, $[|B|+1, |B|]$, $[B-A+1, |B|]$, $\widehat{I}(\alpha)$

$$\frac{1}{\sqrt{q}} \frac{e^{2\pi i \alpha A/q} (1 - e^{2\pi i \alpha (B-A+1)/q})}{1 - e^{2\pi i \alpha/q}} \quad (d) \text{ pf: 其中 } L = \widehat{B} - \widehat{A} + 1, \text{ LHS} = \frac{1}{\sqrt{q}} \sum_{\alpha \bmod q} \frac{|1 - e^{2\pi i \alpha L/q}|}{|1 - e^{2\pi i \alpha/q}|} \leq \sum_{\alpha} \frac{1}{\sqrt{q}} \frac{2}{|1 - e^{2\pi i \alpha/q}|} = \frac{1}{\sqrt{q}} \sum_{\alpha \bmod q} \frac{1}{|\sin(\frac{\pi \alpha}{q})|} \ll \frac{1}{\sqrt{q}} \sum_{\alpha=1}^{(q+1)/2} \frac{\pi}{2\alpha \pi/q} \ll \sqrt{q} \log q, \quad (\text{利用 } x \in [0, \pi/2], \sin x > \frac{2}{\pi} x \text{ 和 } \sin x = \sin(\pi - x)) \quad (e) \text{ pf: 见讲义 2.6.9.}$$

Exercise 7. p 素数, $(\frac{p}{p})$ 是 D 特征模 p , 求证模 p 的最小非二次剩余大小为 $O(\sqrt{p} \log p)$. pf: $\chi(n) := (\frac{n}{p})$ 不是模 1 本原特征, 而是模 p 本原特征. 若 N 是最小的模 p 非二次剩余, 则 $N - 1 = \sum_{n=1}^{N-1} \chi(n) \ll \sqrt{p} \log p$, 故 $N \ll \sqrt{p} \log p$.

Exercise Sheet 6

Exercise 2. (a) $\operatorname{Re}(s) > 1$, $\zeta(s) = \frac{1}{s-1} + i-s \sum_{j=1}^{\infty} \frac{(-1)^j}{j!} \frac{d^j}{ds^j} \zeta(s)$ (给出 $\operatorname{Re}(s) > 0$ 的亚纯延拓); (b) 洛朗展开 $\zeta(s) = \frac{1}{s-1} + \gamma + O((s-1)(s-1))$, $\gamma = \lim_{n \rightarrow \infty} (O(\sum_{k=1}^n \frac{1}{k} - \log n))$. pf. (a) Euler-Maclaurin (讲义 2.2.3), $\zeta(s) = \lim_{n \rightarrow \infty} (1 + \sum_{1 \leq n \leq N} \frac{1}{n^s}) = 1 + \int_1^{\infty} \frac{1}{x^s} dx - \int_1^{\infty} (x-z-\frac{1}{2}) \frac{1}{x^{s+1}} dx + \lim_{n \rightarrow \infty} \frac{(N)^{1-s}/2 - (1)^{1-s}/2}{s} = 1 + \frac{1}{s-1} - s \int_1^{\infty} \frac{1}{x^{s+1}} dx$, (b) $\zeta(s) = \frac{1}{s-1} + (1-\frac{1}{s}) \frac{d}{ds} \zeta(s) + (1+\frac{1}{s}) \frac{d^2}{ds^2} \zeta(s)$, 由讲义 2.2.4, $\gamma = 1 - \int_1^{\infty} \frac{1}{x^2} dx$, 而且 $\int_1^{\infty} \frac{1}{x^2} dx = s \int_1^{\infty} \frac{1}{x^{s+1}} dx = \int_1^{\infty} (\frac{x}{x^2} - \frac{1}{x^{s+1}}) dx = (s-1) \int_1^{\infty} \frac{1}{x^{s+1}} dx = \int_1^{\infty} (x^{s-1} - 1) dx = O(s-1) = \int_1^{\infty} O(\frac{1}{x^2}) O((s-1)) \log x dx = O((s-1)(s-1))$.

Exercise 3. $\prod_{p \leq x} (1 - \frac{1}{p})^{-1} = C \log x \cdot O(1)$. 求证 $C = e^\gamma$. 解: $p f(p) = \prod_{p \leq x} (1 - \frac{1}{p})^{-1}$, $L = \log x$. 由 6-2 (b) 得 $\zeta(1 + \frac{1}{p}) = L + O(1)$, 而且 $\log \zeta(1 + \frac{1}{p}) = \sum_p -\log(1 - p^{-1-\frac{1}{p}}) = \sum_p \sum_{k \geq 1} \frac{e^{-k(1+\frac{1}{p})}}{k} = \sum_p -\log(1 + \frac{1}{p}) + o(1) \ (x \rightarrow \infty)$. 由 Abel, $\sum_p -\log(1 - \frac{1}{p})^{1/L} = \int_0^\infty \log y^{-1/L} d \sum_{p \leq y} -\log(1 - \frac{1}{p}) = \frac{1}{L} \int_2^x y^{-1/L-1} d \sum_{p \leq y} -\log(1 - \frac{1}{p})$. 由 Abel, $\sum_p -\log(1 + \frac{1}{p}) = \frac{1}{L} \int_2^x y^{-1/L-1} d \sum_{p \leq y} -\log(1 + \frac{1}{p}) + o(1)$. 而且 $C \log y + O(1) = C \log y + O(1) = C \log y + O(1) = C \log y + O(1)$, 故上式 $= \frac{1}{L} \int_2^x y^{-1/L-1} [\log C + \log y + o(\frac{1}{\log y})] dy + o(1) = \log L + \log C - \gamma + o(1)$ (换元 $\frac{dy}{y} = d(\log y)$), 所以 $\log C = \gamma$, $C = e^\gamma$.

Exercise 4. χ mod q , $\chi \neq \chi_0$, $\chi \geq 2$. 证明 (a) $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = -L(1, \chi) + O\left(\frac{x}{\sqrt{x}}\right)$, (b) $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = L(1, \chi) \sum_{m \leq x} \frac{\chi(m)}{m} + O\left(\frac{1}{2} \sum_{m \leq x} \chi(m)\right)$, 从而有 $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = O\left(\frac{1}{x}\right)$; (c) $\sum_{n \leq x} \frac{\chi(n) \log p}{p} = \frac{b(x)}{x} + O\left(\frac{1}{\sqrt{x}}\right)$, 其中 $b(x) = \log L(1, \chi) - \sum_{p \leq x} \frac{\chi(p)}{p}$, pf: (a) $\sum_{n=1}^N \frac{\chi(n)}{n} = \frac{S(N)}{N} + s_f \int_{1/N}^N \frac{S(y)}{y^2} dy$, $S(y) = \sum_{n \leq y} \chi(n)$ 有界, 故 L_χ 的 $\sigma_c \leq 0$, 讲义 4.10 $\Rightarrow -L'(1, \chi) = \sum_{n \geq 1} \frac{\chi(n) \log n}{n} = \sum_{n \leq x} \frac{\chi(n) \log n}{n} + \sum_{n > x} \frac{\chi(n) \log n}{n}$, $\sum_{n > x} \frac{\chi(n) \log n}{n} \stackrel{\text{Abel}}{=} \lim_{n \rightarrow \infty} \sum_{n \leq x} \chi(n) \log n \sum_{n \leq x} \frac{\chi(n) \log n}{n} - \int_x^\infty \sum_{n \leq x} \chi(n) \log n \frac{1}{t^2} dt \leq \frac{O(x)}{x} + \sum_{n \leq x} \frac{\chi(n) \log n}{n} = \sum_{n \leq x} \frac{\chi(n) \log n}{n} + \sum_{n \leq x} \frac{\chi(n) \log n}{n} = L(1, \chi) \sum_{n \leq x} \frac{\chi(n) \log n}{n} - \sum_{n \leq x} \frac{\chi(n) \log n}{n} = O\left(\frac{1}{\sqrt{x}}\right)$, 且 $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = O\left(\frac{1}{\sqrt{x}}\right)$, (b) $\log A = 1$, $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = \sum_{n \leq x} \sum_{d|n} \frac{\chi(d) \log d}{d} \frac{\chi(n/d)}{n/d}$, $\sum_{d \leq x} \sum_{n \leq x/d} \frac{\chi(n) \log n}{n} = L(1, \chi) \sum_{d \leq x} \frac{\chi(d) \log d}{d} - \sum_{d \leq x} \frac{\chi(d) \log d}{d} = O\left(\frac{1}{\sqrt{x}}\right)$, 且 $\sum_{d \leq x} \frac{\chi(d) \log d}{d} = O\left(\frac{1}{\sqrt{x}}\right)$, $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = (dm) = e) \sum_{n \leq x} \frac{\chi(n) \log n}{n} = \sum_{n \leq x} \frac{\chi(n) \log n}{n} \frac{\chi(d)}{d}$, Abel 求和 $\Rightarrow O\left(\frac{1}{x}\right)$ (c) $\sum_{n \leq x} \frac{\chi(n) \log p}{p} = \sum_{n \leq x} \frac{\chi(n) \log n}{n} - \sum_{p \leq x} \frac{\chi(p) \log p}{p}$, 由 (b) 第一项有界, 第二项 $\sum_{p \leq x} \frac{\chi(p) \log p}{p} = \sum_{p \leq \sqrt{x}} \frac{\chi(p) \log p}{p} + \sum_{p > \sqrt{x}} \frac{\chi(p) \log p}{p} = O\left(\frac{1}{\sqrt{x}}\right)$.

$$\begin{aligned} \text{有界, (d) } \sum_p \sum_{k \geq 1} \frac{\chi(p^k)}{kp^k} &= -\sum_p \log(1 - \frac{\chi(p)}{p}) = \log L(1, \chi), \text{ 故 } \sum_p \frac{\chi(p)}{p} = \log L(1, \chi) - \sum_p \sum_{k \geq 2} \frac{\chi(p^k)}{kp^k}, \text{ 而 } \sum_{p > x} \frac{\chi(p)}{p} \\ &= \sum_{p > x} \frac{\chi(p) \log p}{p} \frac{1}{\log p} \stackrel{\text{Abel}}{=} \end{aligned}$$

Exercise 5. $|\arg s| \leq \pi - \delta$. 求证 $\Gamma'(s)/\Gamma(s) = \log s - \frac{1}{2s} + O_\delta(|s|^{-2})$. pf: $F(s) \triangleq \log \Gamma(s) - (s - \frac{1}{2}) \log s + s - \frac{\log(2\pi)}{2}$, 由 Stirling, $F(s) = O_\delta(|s|^{-1})$ ($s \rightarrow \infty$, $|\arg s| \leq \pi - \frac{\delta}{2}$). $|s|$ 充分大时, $\exists C_\delta > 0$, $D(s, C_\delta|s|) \subset \{|\arg z| \leq \pi - \frac{\delta}{2}\}$, 故 Cauchy 积分公式 $\rightarrow |F'(s)| \leq \frac{C_\delta}{|s|^2} \sup_{|z-s|=C_\delta|s|} |F(z)| = O_\delta(\frac{1}{|s|^2})$. 故 $F'(s) = O_\delta(\frac{1}{|s|})$.
即分为

Exercise Sheet 7

引理: f 整函数, 非多项式, $f(z) = \sum_{n=0}^{\infty} a_n z^n$, 则 f 阶为 $\rho(f) = \limsup_{n \rightarrow \infty} \frac{\log |a_n|}{\log |a_{n+1}|}$ ($a_n = 0$ 时均按视为 0), p.1: Cauchy 积分公式 $\Rightarrow |a_n| \leq \sup_{|z|=r} |f(z)| \cdot \frac{1}{r^n}$. 若 $\sup_{|z|=r} |f(z)| \leq \exp(r^{\rho+\epsilon})$, 取 $r = n^{\frac{1}{1+\rho+\epsilon}}$, 则 $r \sim n^{\frac{1}{1+\rho+\epsilon}}$, $\log n + O(1) \leq \log n + O(\frac{1}{r}) \leq \log n + O(\frac{1}{n^{\frac{1}{1+\rho+\epsilon}}}) \leq \rho + \epsilon$. 另一方面, 若 $-\log |a_n| \geq \frac{n \log n}{\mu}$, 则 $|a_n| r^n \leq \exp(|z| \log r - \frac{n \log n}{\mu})$, $n \sim r^{\mu} \Rightarrow \sup_{|z|=r} |f(z)| \leq \exp(Cr^{\mu})$ (r 充分大), $\Rightarrow \rho(f) \leq \mu$, $\Rightarrow \rho(f) \leq \limsup_{n \rightarrow \infty} \frac{\log |a_n|}{\log |a_{n+1}|}$.

Exercise 1. $\forall \alpha > 0$, $f_\alpha(z) = \sum_{j=1}^{\infty} \frac{z^j}{\Gamma(1+j/\alpha)}$ 为 α 阶整函数. pf: $a_j \triangleq \Gamma(1+j/\alpha)^{-1}$, Stirling $\Rightarrow -\frac{1}{j} \log |a_j| = \frac{1}{\alpha} (\log j + O(1)) \rightarrow +\infty$, 故收敛半径 $R = \frac{1}{\limsup_{j \rightarrow \infty} |a_j|^{1/j}} = +\infty$, 故整. $\limsup_{n \rightarrow \infty} \frac{n \log n}{-\log |a_n|} = \alpha$, 故阶为 α .

Exercise 2. $f(z) \triangleq \sum_{n=0}^{\infty} e^{-n^2} z^n$ 是非多项式 0 阶整函数. pf: 因为 $\limsup_{n \rightarrow \infty} \frac{-\log n}{-\log |e^{-n^2}|} = 0$.

Exercise 3. 整函数 f 阶为 α , $f(0) \neq 0$, $\forall \epsilon > 0$, $\sum_{n=0}^{\infty} \frac{1}{|f(n)|} < \infty$ (不计 $n=0$). pf: Thm 10.2.6 设 $\sum_{n=0}^{\infty} \frac{1}{|f(n)|} \leq B\alpha + \delta$ 设取点最小格长为 r , 则 $\sum_{n=0}^{\infty} \frac{1}{|f(n)|} < \infty$.

$$\sum_{n>r} \sum_{2^n < |\rho| \leq 2^{n+1}} \frac{1}{|\rho|^{1+\varepsilon}} \ll \sum_{n>r} \frac{\sum_{|\rho| \leq 2^{n+1}} \frac{1}{2^{n(\alpha+\varepsilon)}}}{2^{n(\alpha+\varepsilon)}} \leq \sum_{n>r} \frac{2^{(n+1)(\alpha+\varepsilon)}}{2^{n(\alpha+\varepsilon)}} = \sum_{n>r} 2^{\alpha+\delta-n(\varepsilon-\delta)} < \infty \quad (\text{等比级数, 一开始给 } \delta < \varepsilon),$$

Exercise 4. Hadamard $\zeta^*(s) = e^B \prod_{\rho} (1 - \frac{s}{\rho}) e^{\rho/s}$ on $B = -\sum_{\rho} \frac{1}{\rho} = -2 \sum_{\gamma>0} \frac{\beta}{\beta^2 + \gamma^2}$, 其中 $\rho = \beta + i\gamma$, $\sum_{\rho} \frac{1}{\rho} \stackrel{\text{lim}}{\rightarrow} -\infty$ on $|\rho| < \frac{1}{\rho}$. *Solution.* pf: $\frac{\zeta^*(s)}{\zeta^*(\bar{s})} = B + \sum_{\rho} (\frac{1}{s-\rho} + \frac{1}{\bar{s}-\bar{\rho}})$, $s \rightarrow 0 \rightarrow \frac{\zeta^*(s)}{\zeta^*(\bar{s})} (0) = B$, 由 $\zeta^*(s) = \zeta^*(1-s)$, $\frac{\zeta^*(s)}{\zeta^*(\bar{s})} (0) = -\frac{\zeta^*(s)}{\zeta^*(\bar{s})} (1) = -[B + \sum_{\rho} (\frac{1}{1-\rho} + \frac{1}{\bar{1}-\bar{\rho}})]$, 由 ρ 零点, $1-\rho$ 也为 ζ^* 零点, 故对称主值下 $\sum_{\rho} \frac{1}{1-\rho} = \sum_{\rho} \frac{1}{\rho}$, 得 $B = -\sum_{\rho} \frac{1}{\rho}$. 又 $\rho = \beta + i\gamma$ 是零点时 $\bar{\rho} = \beta - i\gamma$ 也是零点, 故 $B = -\sum_{\gamma>0} (\frac{1}{\beta+i\gamma} + \frac{1}{\beta-i\gamma}) = -2 \sum_{\gamma>0} \frac{\beta}{\beta^2 + \gamma^2}$.

Exercise 10. $n \in \mathbb{N}$. 设 χ 是模 q 的本原 Dirichlet 特征, $\chi(-1) = 1$, 证明 $L(s, \chi) = \sum_{n \in \mathbb{N}} \chi(n) n^{-s} / \prod_{p|q} (1 - \chi(p) p^{-s})$ 解析延拓到整个复平面, 并满足 $\chi(-1)^{-s} q^{(1-s)/2} \Gamma(\frac{s+1}{2}) L(1-s, \bar{\chi}) = \frac{2\pi i}{q} n^{-s/2} q^{s/2} \Gamma(\frac{s}{2}) L(s, \chi)$, 其中 $\bar{\chi}(n) := \sum_{d|(n, q)} \chi(d) \chi(n/d)$; 并证明 $L(s, \chi) = -1$ 时哪里出问题. 解: 除 $q = 1$ 的主特征外, 本原特征非主, 以下看非主特征. 定义 $\theta(x, \chi) := \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2/q}$, 由 $\chi(-1) = 1$, 有 $\theta(x, \chi) = 2 \sum_{n \geq 1} \chi(n) e^{-\pi n^2/q}$. 当 $\text{Re } s > 1$ 时逐项积分, 且 $\int_0^\infty e^{-\pi n^2/q} x^{s-1} dx = (q/\pi)^{s/2} \Gamma(\frac{s}{2}) n^{-s}$, 故记 $I(s, \chi) := \int_0^\infty \theta(x, \chi) x^{s-1} dx$, 则 $I(s, \chi) = 2\pi^{-s/2} q^{s/2} \Gamma(\frac{s}{2}) L(s, \chi)$. 接着按 $n = a + mq$ 并对应内层 Poisson 求和: $\theta(x, \chi) = \sum_{a \bmod q} \chi(a) \sum_{m \in \mathbb{Z}} e^{-\pi(a+mq)^2/q}$, 由 $\sum_{m \in \mathbb{Z}} e^{-\pi(a+mq)^2/q} = \frac{1}{q} \sqrt{\frac{q}{x}} \sum_{k \in \mathbb{Z}} e^{-2\pi i k a/q} e^{-\pi k^2/(qx)}$; 又 $\sum_{a \bmod q} \chi(a) e^{-2\pi i k a/q} = \chi(k) \chi(x)$, 于是 $\theta(x, \chi) = \frac{\chi(x)}{q} \sum_{k \in \mathbb{Z}} \frac{1}{x} \frac{1}{\sqrt{x}} e^{-\pi k^2/(qx)}$, 指数衰减. 由变换式 $x \rightarrow 0^+$ 时也足够衰减, 故 $I(s, \chi)$ 全纯; 再由 (*) 定义 $L(s, \chi) = \frac{1}{2} \pi^{-s/2} q^{-s/2} I(s, \chi) / \Gamma(\frac{s}{2})$, 得到解析延拓. 对 $I(1-s, \bar{\chi})$ 作 $x = 1/y$, 并用上面对 $\bar{\chi}$ 的变换式, 得 $I(1-s, \bar{\chi}) = \frac{1}{\sqrt{q}} I(s, \chi)$; 把 (*) 分别用于 χ 和 $(1-s, \chi)$ 即得函数方程. 若 $\chi(-1) = -1$, 则 $\theta(x, \chi)$ 中 n 与 $-n$ 两项抵消, 所以 $\theta(x, \chi) \equiv 0$, 上面的积分公式不能恢复 $L(s, \chi)$. 应改用 $\theta_1(x, \chi) := \sum_{n \in \mathbb{Z}, n \not\equiv 0 \pmod{q}} \chi(n) e^{-\pi n^2/q}$, Gamma 因子改为 $\Gamma(\frac{s+1}{2})$.

Exercise Sheet 8

Exercise 1. 若存在 $C > 0$ 使 $\zeta(s)$ 任一零- ρ 满足 $\operatorname{Re} \rho < 1 - C/\log(2 + |\operatorname{Im} \rho|)$, 不用显式公式证明存在 $c > 0$ 使 $\psi(x) = \sum_{n \leq x} \Lambda(n) = x + O(xe^{-c\sqrt{\log x}})$. pf: 只需看非整数 x , 整数点由 Perron 半圆公式 $\pm x \pm O(1)$ 逼近. 记 $L = \log(T/2)$, 取 $a = 1 + 1/L$, $b = C/(2\log T)$, $3 \leq T \leq x$, 则 $|\gamma_j| \leq T$, 则零- ρ 点 $\beta + i\gamma$ 满足 $b < 1 - C/(2\log T) \leq 1 - C/(2\log x)$ 故 $b < 1$ 取小取常数, 故形如 $b \leq \operatorname{Re} s \leq a$, $|\operatorname{Im} s| \leq T$ 中 $-C'/C$ 只有 s 的简单极点, 且 $C'/C(s) = 1/(8 + \log T)$. 用 Perron 公式给 $\psi(x) = \frac{1}{2\pi i} \int_{b-i\infty}^{b+i\infty} (-C'/C(s))x^s ds + O(xL^2/T)$, 由柯西留数定理移到 $\operatorname{Re} s = b$, 穿过 $s = 1$ 的留数为 x , 无零形中标准 Jensen-Cauchy 估计给 $-C'/C(s) \ll L^{-2}$, 故新竖线为 $O_C(x^{1/L})$, 横边与截断误差为 $O_C(xL^2/T)$, 于是 $\psi(x) = x + O_C(x^{1/L}L^2 + xL^2/T)$. 取 $T = \exp(\sqrt{A\log x})$, 则 $x^b = x \exp(-CL/(2\log T)) = xe^{-A\sqrt{T}}$ 且 $x/T = xe^{-A\sqrt{T}}$; 吸收对数幂得 $\psi(x) = x/T + O(xe^{-c_1\sqrt{\log x}})$.

Exercise 2. 设 μ 为 Mobius 函数, 用 Perron 公式证明存在 $c > 0$ 使 $\sum_{n \leq x} \mu(n) = O(xe^{-c\sqrt{\log x}})$, 可用事实是: 若 $\zeta(\sigma + it) \neq 0$ 对 $\sigma \geq 1 - c_0 / \log(|t| + 2)$ 成立, 则当 $\sigma \geq 1 - \frac{c_0}{\log(|t| + 2)}$ 时 $1/\zeta(\sigma + it) \ll_{c_0} \log(|t| + 2)$. pf: 设非零区域为 $\zeta(\sigma + it) \neq 0$, $\sigma \geq 1 - \kappa / \log(|t| + 2)$. 由题中 fact, 若 $\sigma \geq 1 - \kappa / \log(|t| + 2)$, 则 $1/|\zeta(\sigma + it)| \ll_{\kappa} \log(|t| + 2)$. 记 $M(x) = \sum_{n \leq x} \mu(n)$, $L = \log x$, $T = \log(T + 2)$, 取 $a = 1 + 1/L$, $b = 1 - \kappa / (2 \log T)$, Perron 公式给 $M(x) = \frac{1}{2\pi i} \int_{-iT}^{+iT} x^s \zeta(s) (\frac{x}{L})^{s-1} ds + O(x/LT)$. 移到 $\text{Re } s = b$: 因 $1/\zeta(s)$ 在矩形内全纯且 $s = 1$ 是 L/T 的零点而非极点, 取 α 为无理数. 由 fact, 竖线为 $O_{\kappa}(x/LT)$, 横边与截断误差为 $O_{\kappa}(x/LT)$, 故 $M(x) = O_{\kappa}(x^b L_T^{\frac{1}{2}} + x/LT)$. 取 $T = \exp(b\sqrt{L})$, $B = \sqrt{\kappa/2}$, 则 $x^b = xe^{-B\sqrt{L}}$ 且 $x/T = xe^{-B\sqrt{L}}$, 吸收对数得 $\sum_{n \leq x} \mu(n) = O(xe^{-c_2\sqrt{\log x}})$.

Exercise 3. $N \in \mathbb{N}$, $\text{Li}(x) = \frac{x}{\log x} \sum_{n=0}^{N-1} \frac{n!}{(\log x)^{n+1}} + O_N\left(\frac{x}{(\log x)^{N+1}}\right)$. pf: $\text{Li}(x) = \int_2^x \frac{1}{\log u} du$, 反复分部积分得 $\frac{x}{\log x} \sum_{n=0}^{N-1} \frac{n!}{(\log x)^{n+1}} + \int_2^x \frac{du}{(\log u)^{N+1}}$. 余项 $= \int_2^{\sqrt{x}} \frac{du}{(\log u)^{N+1}} + \int_{\sqrt{x}}^x \frac{du}{(\log u)^{N+1}}$

Exercise 4. 记 $M(x) := \sum_{n \leq x} \mu(n)$ 与 $\psi(x) := \sum_{n \leq x} \Lambda(n)$, 则在 $0 \leq \theta \leq 1$ 下, $M(x) = O_e(x^{1-\theta+\varepsilon})$ 当且仅当 $\psi(x) = x + O_e(x^{1-\theta+\varepsilon})$. pf: $\theta = 0$ 平凡, 设 $0 < \theta \leq 1$, 置 $\alpha = 1 - \theta < 1$. 先证两边都等价于 $\zeta(s) \neq 0$ 对 $\text{Res} > \alpha$. 若 $M(x) = O_e(x^{\alpha+\varepsilon})$, 则 $\text{Res} > 1$ 时 $1/\zeta(s) = \sum \mu(n)n^{-s} = \int_1^{\infty} M(t)t^{s-1} dt$; 对 $\sigma = \text{Res} > \alpha$

取 $\eta > 0$ 使 $\alpha + \eta < \sigma$, 被积函数为 $O((\eta\sigma)^{\sigma-1})$, 故积分在 $Res > \alpha$ 局部一致收敛, 给出 $1/\zeta$ 全纯延拓, 于是 ζ 无零. 若 $\psi(x) = x O(x^{\alpha+\eta})$, 当 $\psi = x + E$, 则 $Res > 1$ 时 $-\zeta'/\zeta(s) = s \int_0^\infty \psi(x) x^{s-1} dt = \alpha \int_0^\infty x^\alpha + s \int_0^\infty E(t) x^{s-1} dt$, 同理后一项积分在 $Res > \alpha$ 全纯, 故 $-\zeta'/\zeta(s) - 1/(s-1)$ 全纯, 若在此半平面有极点, 则 $-\zeta'/\zeta$ 有极点, 矛盾. 反过来设 $\zeta(s) \neq 0$ 对 $Res > \alpha$, 取 $\eta = \alpha/2$, 取 $b = \alpha + \zeta/s$, $c = 1 + 1/\log x$, $T = x$, 在 $Res > b$ 中标准零点避开估计 $1/(s-\zeta) \ll |t| + 2/\zeta$ 与 $-\zeta'/\zeta(s) \ll \log^2(|t| + 3)$, Perron 公式给 $M(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{e^{xs}}{s-\zeta(s)} ds + O(\log x)$, 移线到 $Res = b$, 不穿过留数, 新紧线 $\ll_c x^{b/T} (T + 2)^{1/3} (1+1)^{-1} dt = O_c(x^{\alpha+\eta})$, 横边与截断差可忽略, 故 $M(x) = O_c(x^{\alpha+\eta})$. 同样 $\psi(x) = x O(x^{\alpha+\eta})$, 移线只穿过 b , 因 $-\zeta'/\zeta(s) - 1/(s-1) = O_c(x^{\alpha+\eta})$, 留数为 $Res_{s=1}((-\zeta'/\zeta(s))x^s/s) = x$. 类似下降线 $\ll_c x^{b/T} \log^2(t+3)(1+1)^{-1} dt = O_c(x^{\alpha+\eta})$. 故 $\psi(x) = x O(x^{\alpha+\eta})$. 整数 x 外用 Perron 半圆公式 $x + O(1)$ 可证即可, 代回 $\alpha = 1 - \theta$ 得证.

命题 8.0.1 (柯西积分公式 (Cauchy integral formula)) 设 $\Omega \subset \mathbb{C}$ 为区域, f 在 Ω 上全纯. 若 $\overline{D(z_0, r)} \subset \Omega$, 则对任意 $n \geq 0$, $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|\zeta - z_0|=r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$.

留数定理 (residue theorem). 设 γ 为正向简单闭曲线, f 在 γ 及其内部除有限个孤立奇点 $a_1, \dots, a_m$ 外全纯, 则 $\int_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^m \text{Res}(f, a_j)$.